

PRACTICAL OSSP EXPERIMENTS – 2

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Program-1:

(Cmd executor code program)

Code:

```
GNU nano 8.7.1
#include <stdio.h>
#include <unistd.h>
#include <sys/types.h>
#include <sys/wait.h>
#include <stdlib.h>

int main() {

    char cmd[100];

    printf("Enter Linux command: ");
    scanf("%s", cmd);

    pid_t pid = fork();

    if (pid == 0) {
        printf("Child PID : %d\n", getpid());
        execlp(cmd, cmd, NULL);
        perror("Execution Failed");
    }

    else if (pid > 0) {
        printf("Parent PID : %d\n", getpid());
        wait(NULL);
        printf("Child process completed.\n");
    }

    else {
        printf("Fork failed\n");
    }

    return 0;
}
```

Output:

```
hersh@Ubuntu:~/C programs$ nano cmd.c
hersh@Ubuntu:~/C programs$ ./cmd
Enter Linux command: ls
Parent PID : 6081
Child PID : 6085
childParentPID    cmd    cp      cp_prog.c  fork.c  getID.c  main.c  name.c  note1.txt  prac2.c  wait.c  welcome.c
childParentPID.c  cmd.c  cp_prog  fork      getID   main     name    note.txt  prac2     wait     welcome
Child process completed.
hersh@Ubuntu:~/C programs$
```

Experiment – 2:

Copy content file from “note.txt” to note1.txt”

Code:

```
GNU nano 8.7.1                                     prac2.c
#include <stdio.h>
#include <fcntl.h>
#include <unistd.h>

int main() {
int src, dest;
char buffer[1024];
int bytes;

src = open("note.txt", O_RDONLY);

if (src < 0) {
printf("Cannot open source file\n");
return 0;
}
dest = open ("note1.txt", O_WRONLY | O_CREAT | O_TRUNC, 0644);
while ((bytes = read(src, buffer, sizeof(buffer))) > 0) {
write(dest, buffer, bytes);
}
close(src);
close(dest);
printf("File copied succesfully!\n");
}
```

Output:

```
hershy@Ubuntu:~/C programs$ nano prac2.c
hershy@Ubuntu:~/C programs$ gcc prac2.c -o prac2
hershy@Ubuntu:~/C programs$ ./prac2
bash: ./prac2: No such file or directory
hershy@Ubuntu:~/C programs$ ./prac2
Cannot open source file
hershy@Ubuntu:~/C programs$ nano prac2.c
hershy@Ubuntu:~/C programs$ gcc prac2.c
hershy@Ubuntu:~/C programs$ ./prac2
Cannot open source file
hershy@Ubuntu:~/C programs$ nano prac2.c
hershy@Ubuntu:~/C programs$ gcc prac2.c
hershy@Ubuntu:~/C programs$ ./prac2
File copied succesfully!
hershy@Ubuntu:~/C programs$ cat note1.txt
Hello I'm harshit Sajjanapu

hershy@Ubuntu:~/C programs$ cat note.txt
Hello I'm harshit Sajjanapu

hershy@Ubuntu:~/C programs$ █
```